

systems in which each component operates with minimal, clearly defined privileges—a fundamental element of our cyber-immune approach. We structure the system into parts based on their focus—security, functionality, or business logic. The microkernel establishes a ‘trusted execution area’ for essential components, isolating them from threats. This lets us prioritise the protection of key modules while enabling other parts to operate independently, creating a trusted system at a reduced cost.

Q. What is the key security difference between embedded operating systems and general-purpose ones?

A. A general-purpose operating system is designed to run a wide range of software on computers, whether in corporate or consumer environments. The underlying hardware assumes that extra security layers can be added, shifting the focus to supplementary software such as antivirus and network threat protection.

In contrast, IoT systems are purpose-built, designed for operational simplicity and resource efficiency. These systems should ideally be secure by design rather than depending on external countermeasures like security gateways. The fundamental difference is that IoT systems must be secure from the beginning.

Q. How do you balance security with performance and power in embedded designs?

A. In IoT design, trade-offs are inevitable, but we consider security to be just as critical as functionality or business requirements. People rarely question the balance between power and business needs, yet often challenge the trade-off between security and performance. If we treat security as equally vital, then we design the system to meet all requirements simultaneously: functionality, business

logic, and security.

Q. Which industries benefit the most from embedded security?

A. Sectors with stringent security demands—such as transportation, industrial control systems, building automation, and the Internet of Things (IoT)—benefit significantly from embedded security. These industries depend on systems that manage sensitive data and control essential operations, making them high-value targets for cyberattacks. By embedding security directly into the system architecture through our cyber-immune framework, we ensure that even in the event of an attack, the impact is contained and critical operations are safeguarded. This is especially important for industrial gateways, IoT devices, and control systems where reliability and resilience are non-negotiable.

Q. How do security needs differ between consumer IoT and critical systems like energy or telecom?

A. While the architectural approach remains consistent across systems, the required security levels and focus areas differ. In critical infrastructure—such as nuclear power or telecommunications—the emphasis is on ensuring system availability and preventing service disruptions, as even minor breaches can lead to serious consequences. Consumer IoT, on the other hand, prioritises data confidentiality and user privacy, with a comparatively higher tolerance for risk. The security implementation varies in intensity, based on the specific risk profile and operational importance of each use case.

Q. Does open source play a role in embedded security?

A. Open source is an effective way to engage the community and enhance solutions. For us, it is more

than outreach—we actively release components of our code, gradually increasing transparency. As security experts, we reject the idea of ‘security through obscurity’. We have already published portions of our code and invited the open source community to collaborate. We also offer a free KasperskyOS Community Edition, allowing developers, students, and hobbyists to explore our technology and build non-commercial projects.

Q. How does open source fit with Kaspersky and embedded systems?

A. Open source strengthens collaboration and promotes innovation in IoT systems. We are committed to sharing our code in stages and encouraging contributions from the open source community. Our philosophy is grounded in transparency—security should not depend on keeping things hidden. By offering the KasperskyOS Community Edition, we give developers and learners access to our tools to build and experiment securely.

Q. How do you address the security risks of open source?

A. Building secure systems—whether open source or proprietary—requires expertise. What truly matters is the competence of those creating the systems. We support and mentor the developer community by sharing our extensive knowledge of cybersecurity. Having delivered security solutions since the previous century, Kaspersky now brings this legacy to both students and open source contributors. **END** 🐼

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